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3D Modeling of Complex Hydraulic Fracture Geometry from the Kamioka Field Test

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Abstract

This paper presents a high-fidelity 3D hydraulic fracture model based on the phase-field approach, following a field pilot test conducted at the Kamioka mine in central Japan. The phase-field method represents fractures using a continuous variable that transitions from 0 (fractured) to 1 (intact). This approach eliminates the need for computational meshes to conform to discontinuous fracture planes. This method enables efficient tracking of complex hydraulic fracture geometry evolution in three dimensions, particularly when modeling interactions with natural fractures. The simulated fracture morphology shows non-planar propagation interacting with pre-existing natural fractures. Our numerical approach captures these complex fracture behaviors, which have also been observed in the field data. Analysis of microseismic event locations relative to observed fracture geometry enhances understanding of the connection between seismic activity and physical fracture formation. This study advances understanding of how complex 3D hydraulic fractures interact with natural discontinuities. The findings enhance predictive capabilities for reservoir stimulation design in complex geological settings where conventional planar fracture models may be inadequate. This integrated approach combines numerical simulation with field-inspired modeling. It offers valuable insights for improving hydraulic fracturing operations in subsurface engineering applications.

Introduction

Hydraulic fracturing is essential for economically viable production from unconventional reservoirs. Its success hinges critically on accurately predicting fracture network development in heterogeneous subsurface conditions. While planar 2D models remain industry standard for operational planning, field evidence consistently demonstrates that actual fracture geometry exhibits complex 3D behavior. This complexity is particularly pronounced when interacting with natural fractures and bedding planes (Warpinski et al., 2009; Fisher et al., 2012). This discrepancy often leads to suboptimal stimulation design.

The modeling challenge becomes particularly acute when pre-existing natural fractures interact with induced hydraulic fractures. These interactions can potentially cause deflection, branching, arrest, or complex network development (Dahi-Taleghani and Olson, 2011). Conventional numerical methods for fracture simulation face fundamental limitations in representing these complex behaviors. These approaches include discrete fracture networks, finite element methods with cohesive zones, and extended finite element methods. They typically require explicit fracture surface representation, complex remeshing procedures, and struggle with handling branching or coalescence phenomena efficiently (Carrier and Granet, 2012; Mohammadnejad and Khoei, 2013).

Traditional fracture modeling approaches present distinct limitations for complex geological environments like Kamioka. Boundary element methods excel at single planar fractures but struggle with branching geometries. Discrete fracture network models require pre-defined fracture geometries. Extended finite element methods demand complex remeshing and struggle with multiple interacting fractures. In contrast, the phase-field approach accommodates all these complex behaviors within a unified framework. This makes it particularly suitable for metamorphic terranes like Kamioka where fracture-foliation interactions dominate the hydraulic fracturing process.

The phase-field approach offers a compelling alternative for modeling complex fracture propagation. This method represents fractures using a continuous field variable that transitions smoothly from fractured (0) to intact material (1). It regularizes sharp crack discontinuities within a diffuse interface framework (Bourdin et al., 2012). By eliminating the need for explicit fracture tracking or remeshing, the phase-field method accommodates complex fracture behaviors. These include branching, merging, and non-planar propagation, all within a unified mathematical formulation that remains computationally tractable for field-scale problems.

For the Kamioka field test specifically, the phase-field approach offers critical advantages: (1) natural handling of fracture interactions with pre-existing metamorphic foliation without requiring prior knowledge of natural fracture orientations; (2) seamless tracking of non-planar fracture propagation observed in the fluorescent resin cores; (3) unified treatment of fracture branching and arrest mechanisms that conventional methods struggle to capture; (4) direct correlation capability with diffuse microseismic event distributions rather than discrete fracture surfaces.

Recent advances in phase-field modeling have demonstrated promising capabilities for hydraulic fracturing simulation. Wheeler et al. (2014) extended the approach to porous media with fixed pressure fields. Mikelic et al. (2015) developed it further for variable pressure conditions. You and Yoshioka (2023) proposed a micromechanically derived phase-field model with diffused poroelasticity that more accurately represents fracture apertures and fluid-solid coupling. However, a critical research gap remains: most phase-field applications have focused on 2D problems or idealized 3D scenarios with limited validation against field-scale observations in geologically complex environments.

This paper addresses two primary objectives: (1) to present a robust 3D hydraulic fracture model based on the phase-field approach that efficiently captures complex fracture geometry evolution; (2) to provide a benchmark case for future modeling studies on hydraulic fracture propagation in heterogeneous media with pre-existing discontinuities. For practicing engineers, this work offers several practical applications. These include improved prediction of stimulated reservoir volume in complex geology, better assessment of fracture containment risks, and more accurate production forecasting from stimulated wells. This modeling approach can inform completion design optimization and well spacing decisions in reservoirs with significant natural fracture systems.

The paper proceeds as follows: First, we present the theoretical framework of the phase-field approach for hydraulic fracturing, including governing equations and numerical implementation. Next, we describe the Kamioka field test setup and key observations. We then detail our numerical modeling approach and simulation results. Finally, we discuss the implications for hydraulic fracturing operations and outline directions for future research and model enhancement.

Statement of Theory and Definitions

Phase-field model formulation

The phase-field approach provides a robust mathematical framework for modeling complex fracture propagation without explicit tracking of discontinuities. This section presents the theoretical foundation of our modeling approach for hydraulic fracturing in poroelastic media.

The phase-field model for fracture is derived from the variational approach to fracture mechanics proposed by Francfort and Marigo (1998). In this formulation, fracture propagation is treated as an energy minimization problem. For a domain $\Omega \subset R^3$ containing a lower-dimensional fracture set Γ (Fig. 1), the total energy of the system is represented as:

$$E(\boldsymbol{u}, \Gamma) := \int_{\Omega \setminus \Gamma} \psi(\boldsymbol{u}) dV + \int_{\Gamma} G_c dS$$

where u is the displacement field, $\psi(\boldsymbol{u})$ is the strain energy density, and G_c is the critical energy release rate or fracture toughness.

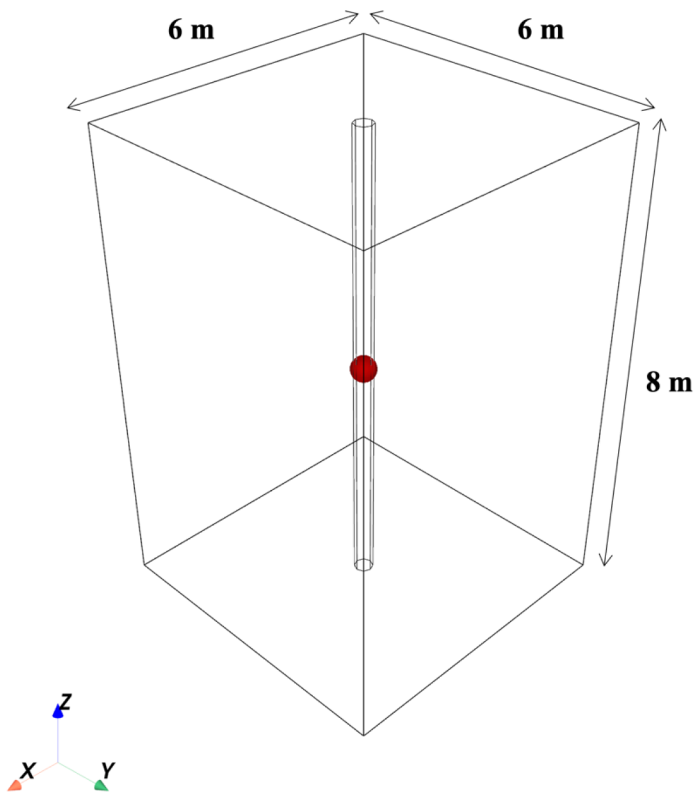

Figure 1—Three-dimensional wireframe model showing central borehole and injection point for the Kamioka field test simulation.

The phase-field method, following Bourdin et al. (2000), regularizes this energy functional by introducing a continuous scalar field variable $\upsilon \in [0, 1]$, which transitions smoothly from intact material ($\upsilon = 1$) to fully fractured material ($\upsilon = 0$). This regularization is controlled by a length parameter l_s:

$$E_l(\boldsymbol{u}, v) := \int_{\Omega} \psi(\boldsymbol{u}, v) dV + \int_{\Omega} \frac{G_c}{4c_n}\left[\frac{(1-v)^n}{l_s} + l_s|\nabla v|^2\right] dV$$

where c_n is a normalization parameter given by $c_n := \int_0^1 (1-\omega)^{n/2} d\omega$. Two common formulations exist: AT1 for n = 1 and AT2 for n = 2, with AT1 typically providing more accurate crack opening displacements (Yoshioka et al., 2020).

For hydraulic fracturing applications, this energy functional must be extended to account for fluid pressure within fractures and poroelastic effects in the surrounding medium. The total energy functional for a poroelastic medium with hydraulic fracture is given by:

$$F_l(\boldsymbol{u}, v, p) := W(\boldsymbol{u}, v, p) + \int_\Omega \frac{G_c}{4c_n}\left[\frac{(1-v)^n}{l_s} + l_s|\nabla v|^2\right] dV$$

where $W(\boldsymbol{u}, v, p)$ represents the poroelastic strain energy accounting for the effects of fluid pressure p. The poroelastic strain energy formulation builds upon recent advances in micromechanically-consistent phase-field modeling (You and Yoshioka, 2023). Rather than applying damage directly to fluid pressure terms as in earlier approaches, this method modifies the underlying poroelastic coefficients to reflect the physical changes in rock structure during fracturing. This strain energy can be expressed as:

$$W(\boldsymbol{u}, v, p) := \int_\Omega \psi(\boldsymbol{u}, v) dV + \int_\Omega \frac{M}{2}[\alpha(v)\nabla \cdot \boldsymbol{u} - \zeta(v)]^2 dV$$

where $\alpha(v)$ is the phase-field dependent Biot coefficient, $M(v)$ is the Biot modulus, and $\zeta(v)$ is the variation of fluid content.

The degraded elastic strain energy $\psi(\boldsymbol{u}, v)$ accounts for the stiffness reduction due to fracturing and can be formulated using various strain energy decomposition schemes. In this work, we employ the volumetric-deviatoric (V-D) decomposition:

$$\psi_{vd}(\boldsymbol{u}, v) = g(v)\left[\frac{k_m}{2}\langle Tr(\boldsymbol{\varepsilon})\rangle_+^2 + \mu_m \boldsymbol{\varepsilon} : \mathbb{K} : \boldsymbol{\varepsilon}\right] + \frac{k_m}{2}\langle Tr(\boldsymbol{\varepsilon})\rangle_-^2$$

where k_m *and* μ_m are the bulk and shear moduli of the porous matrix, respectively. The Macaulay brackets $\langle \cdot \rangle_\pm = \frac{(|\cdot| \pm \cdot)}{2}$ decompose volumetric strain into tensile (+) and compressive (–) components, addressing the asymmetric nature of fracture behavior under different loading conditions. We select the quadratic degradation function $g(v) = v^2$ for material deterioration, providing numerical stability for brittle failure behavior.

This formulation provides several key advantages for modeling hydraulic fracturing:

1. Complex fracture geometries, including branching and merging, are handled without specialized meshing or explicit tracking algorithms
2. Phase-field evolution occurs through energy minimization, capturing the competition between elastic energy reduction and surface energy increase
3. Poroelastic coupling accounts for the interaction between fluid pressure and mechanical deformation
4. The diffuse representation of fractures eliminates numerical challenges associated with discontinuities

The phase-field variable v and displacement field u evolve to minimize the total energy functional, subject to irreversibility constraints that prevent fracture healing.

Governing equations for poroelastic media

The hydraulic fracturing process observed at the Kamioka field test involves complex interactions between rock deformation, fluid flow, and fracture development within a naturally fractured metamorphic rock mass. To capture these coupled processes numerically, we employ a system of three governing equations that describe the evolution of mechanical displacement, fluid pressure, and fracture damage.

The mechanical response of the rock mass under fluid injection follows quasi-static equilibrium, where inertial effects are negligible. The force balance equation governs the displacement field u:

$$\nabla \cdot \boldsymbol{\sigma} + \boldsymbol{b} = 0$$

where the total stress σ incorporates both elastic deformation and fluid pressure effects according to Biot's effective stress principle. For the phase-field representation of fractures, the constitutive relationship becomes:

$$\boldsymbol{\sigma} = C_{eff}(v) : \boldsymbol{\varepsilon} - \alpha(v) p \mathbf{I}$$

The effective stiffness tensor $C_{eff}(v)$ degrades as fractures develop, transitioning from intact rock properties to those of a fractured medium. Following You and Yoshioka (2023), we adopt a volumetric-deviatoric decomposition to distinguish between tensile and compressive loading conditions:

$$\mathbb{C}_{eff}(v) = 3k_{eff}(v)\mathbb{J} + 2g(v)\mu_m \mathbb{K}$$

where $k_{eff}(v)$ = $[g(v)H(Tr(\boldsymbol{\varepsilon})) + H(-Tr(\boldsymbol{\varepsilon}))]$ km accounts for the different responses of fractures under extension versus compression.

The fluid flow behavior at Kamioka involves both matrix flow through the low-permeability metamorphic rock and enhanced flow through induced fractures. The mass balance equation for the fluid phase is:

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot \boldsymbol{q} = Q$$

where ζ represents the change in fluid content, $\boldsymbol{q}$ is the fluid flux, and Q accounts for injection sources. The fluid flux follows Darcy's law with enhanced permeability in fractured regions:

$$\boldsymbol{q} = -\frac{\boldsymbol{K}(v)}{\mu} \nabla p$$

The fundamental coupling between mechanical and hydraulic processes occurs through the fluid content variation $\zeta = \alpha(v)\nabla \cdot \boldsymbol{u} + \frac{p}{M(v)}$. This expression quantifies how rock deformation affects fluid storage and pressure: volumetric compression expels fluid (increasing pressure), while dilation draws fluid in (reducing pressure). For the Kamioka injection scenario, this coupling mechanism governs both the pressure buildup that drives fracture initiation and the subsequent pressure redistribution as new flow pathways develop through the induced fracture network.

The evolution of fracture damage follows from energy minimization principles, where the phase-field variable v transitions from intact material ($v = 1$) to fully fractured material ($v = 0$). The governing equation for phase-field evolution incorporates both mechanical and hydraulic driving forces:

$$\frac{\delta \mathcal{F}_l}{\delta v} = 2v\psi_e^+ + \frac{p^2}{2}\frac{\partial(1/M)}{\partial v} + \frac{G_c}{4c_n}\left[-\frac{n(1-v)^{n-1}}{l_s} + 2l_s \Delta v\right] = 0$$

This equation balances the reduction in elastic energy, changes in fluid storage capacity, and the energy required to create new fracture surface area. The irreversibility constraint $\dot{v} \le 0$ ensures that fractures cannot heal once formed.

The coupling between mechanical damage and hydraulic properties reflects the physical changes occurring in the Kamioka rock mass during fracturing. The Biot coefficient evolves according to:

$$\alpha(v) = 1 - [g(v)H(Tr(\varepsilon)) + H(-Tr(\varepsilon))](1 - \alpha_m)$$

This relationship captures the transition from matrix-dominated flow ($\alpha_m \approx 0.6$ for the metamorphic rock) to fracture-dominated flow ($\alpha \rightarrow 1$ in open fractures). The corresponding changes in fluid storage capacity follow:

$$1/M(v) = (\alpha(v) - \varphi(v))/k_s + \varphi(v)c_f$$

where effective porosity increases from matrix values to unity in fully developed fractures, reflecting the enhanced storage capacity observed in the resin-filled fracture network.

The anisotropic permeability tensor accounts for enhanced conductivity in the induced fracture network, as evidenced by the resin-filled fractures observed under black light in the recovered cores (Takeuchi et al., 2025). The permeability enhancement follows:

$$\boldsymbol{K}(v) = K_m \boldsymbol{I} + K_{frac}(v)\left(\boldsymbol{I} - \boldsymbol{n}_\Gamma \bigotimes \boldsymbol{n}_\Gamma\right)$$

where the fracture contribution $K_{frac}(v) = (1-v)^{\xi}\frac{w2}{12}$ represents the cubic law for flow in parallel-plate fractures and enables the fluid pathways that allow resin penetration throughout the fracture network. The fracture aperture $w = h_e\ Tr(\boldsymbol{\varepsilon})$ is estimated from the volumetric strain, linking local deformation to hydraulic conductivity, with h_e representing the characteristic element size. The fracture normal direction $\boldsymbol{n}_\Gamma$ is approximated from the principal strain directions, avoiding the numerical complexities associated with computing fracture orientation from phase-field gradients.

The weighting exponent $\xi = 10$ provides sufficient contrast between matrix and fracture permeabilities while maintaining numerical stability. This regularization is essential because setting $w = 0$ everywhere except in fully fractured elements ($\upsilon = 0$) creates excessive permeability contrasts that lead to convergence difficulties.

This system of equations captures the essential physics governing the hydraulic fracturing process observed at Kamioka: injection-induced pressure buildup creates mechanical stress, leading to fracture initiation and propagation, which in turn modifies the flow paths and mechanical properties of the rock mass. The resulting coupled evolution of fracture geometry, stress redistribution, and fluid flow provides the foundation for comparing numerical predictions with the field observations of fracture patterns and microseismic activity.

Numerical implementation

The coupled system of equations governing hydraulic fracturing in poroelastic media presents significant computational challenges due to the strong nonlinear coupling between mechanical deformation, fluid flow, and fracture evolution. To address these challenges while maintaining numerical stability and computational efficiency for field-scale simulations like the Kamioka test, we employ a staggered solution strategy implemented within the open-source finite element framework OpenGeoSys (Bilke et al., 2022).

The numerical solution approach decomposes the three-field problem (displacement u, pressure p, and phase-field υ) into sequential sub-problems that are solved iteratively until convergence. This staggered scheme takes advantage of the convexity properties of the phase-field energy functional with respect to individual field variables while managing the complex coupling between them.

The iterative coupling strategy employs a fixed-stress splitting approach to manage the strong nonlinearity between mechanical deformation and fluid flow (Kim et al., 2011). This method maintains mechanical equilibrium during pressure updates by enforcing $k_{eff}(\upsilon)\boldsymbol{\delta}\colon \varepsilon_{i,j} - \alpha(\upsilon)\ p^{i,j} = constant$ within each iteration cycle. The approach eliminates restrictive time-step limitations while preserving the physical coupling mechanisms.

This constraint leads to a stabilized flow equation that includes both storage and coupling terms:

$$\int_\Omega w_p \frac{1}{M(v)} \frac{p^{i,j} - p^{i-1}}{\Delta t} dV + \int_\Omega w_p \frac{\alpha 2(v)}{k_{eff}(v)} \frac{p^{i,j} - p^{i,j-1}}{\Delta t} dV + \int_\Omega \frac{K}{\mu} \nabla p^{i,j} \cdot \nabla w_p dV$$
$$= \int_\Omega w_p Q d\Omega - \int_{\partial\Omega_q} w_p \bar{\boldsymbol{q}} dS - \int_\Omega w_p \alpha(v) \frac{\delta\left(\varepsilon^{i,j-1} - \varepsilon^{i-1}\right)}{\Delta t} dV$$

The fixed-stress split approach provides stable coupling between mechanical deformation and fluid flow, avoiding the convergence difficulties that can occur with alternative coupling schemes. This stabilization approach eliminates spatial discretization restrictions and enables efficient solution of field-scale problems.

The phase-field variable υ evolves according to the energy minimization principle with irreversibility constraints. The discrete weak form of the phase-field equation becomes:

$$\int_{\Omega} w_v\left[2v\psi_e^{+}+\frac{p^2}{2}\frac{\partial\left(\frac{1}{M}\right)}{\partial v}\right]dV-\int_{\Omega} w_v\frac{G_c}{4c_n}\frac{n(1-v)^{n-1}}{l_s}dV-\int_{\Omega}\frac{G_c}{2c_n}l_s\nabla w_v\cdot\nabla v dV=0$$

To enforce the irreversibility condition ($\dot{\upsilon}\leq 0$) and bound the solution space $\upsilon \in [0, 1]$, we employ the variational inequality solver provided by PETSc. Rather than strict irreversibility, we apply a relaxed constraint where $\upsilon(x, t^{i-1}) \leq \upsilon_{irr}$, with $\upsilon_{irr} = 0.1$ to allow some damage reversibility before complete fracture formation.

The spatial discretization employs isoparametric finite elements with bilinear shape functions for all field variables. This choice, while not satisfying the Ladyzhenskaya-Babuška-Brezzi condition for mixed formulations, is stabilized by the fixed-stress splitting approach. The element size is chosen to satisfy the relationship $l_s/h_e \geq 2$ to ensure adequate resolution of the diffuse fracture interface.

For temporal discretization, we employ a backward Euler scheme for the flow equation to ensure unconditional stability. The time step size is adaptively controlled based on the phase-field evolution rate, with smaller steps during active fracture propagation and larger steps during quasi-static periods. For the Kamioka simulation, small time steps ($\Delta t = 0.001$ s) are employed during the injection phase to capture rapid fracture dynamics.

The implementation takes advantage of the modular structure of OpenGeoSys, allowing different physical processes to be activated or deactivated as needed. For the Kamioka simulation, the full hydro-mechanical-phase-field coupling is active during fracture propagation phases, while simplified flow solutions can be used during periods of quasi-static behavior.

The staggered solution approach requires iteration between the field variables until convergence. Convergence is assessed using relative residual norms with tolerances of 1×10^{-6} for all individual processes (displacement, pressure, and phase-field), reflecting the multi-physics coupling requirements of the hydraulic fracturing simulation.

This numerical framework provides the computational foundation for simulating the complex 3D hydraulic fracturing process, including the interaction between induced fractures and pre-existing natural fractures within the metamorphic rock mass. The robust coupling strategies and adaptive solution procedures enable accurate tracking of fracture evolution while maintaining computational efficiency for field-scale problems.

Description of the Kamioka Field Test

The following detailed description of the Kamioka field test methodology is reproduced verbatim from our prior work (Takeuchi et al., 2025):

Site geology and test setup

The Kamioka field test was conducted inside a branch of the mine drift in the Tochibora pit of the Kamioka Mine in Hida City, Gifu Prefecture, central Japan. The experimental site was set up in an open space in the branch of the mine drift, with area dimensions of 6, 9, and 5 m in width, length, and height, respectively. The overburden at the experimental site was estimated at 465m, based on the experimental site floor was elevated at 520 m and the elevation of the top surface at the site was 985 m. The Kamioka mine is located within the central body of the Hida Metamorphic Belt. Neutral to basic complex rock bodies called "metabasite" are elongated in a north–south trend in this mining area, and folding gneisses are distributed on the wing.

"Inishi rock" which is a type of migmatite, is distributed on the wing of a metabasite complex accompanied by metamorphosed limestone. The major rock types in the Kamioka mine are composed of gneisses, Inishi rock, skarn, and aplite. Open natural fractures were developed in the bedrock of the experimental site.

Experimental procedure

The resin fracturing borehole (RF1) was drilled vertically to a depth of approximately 9 m with a 76 mm diameter core barrel. Five AE observation boreholes (AE4--AE7, plus AE2 reused from preliminary testing), each with a depth of 8 m and diameter of 76 mm, were positioned 1.2 m apart from RF1 and arranged in a circular pattern, rotated approximately 72° around RF1. Two sections at RF1 (center depth of 2.10 and 3.30 m) were selected as fracturing sections to prevent natural fractures in the injection section (475 mm) based on observation of the recovered core. The double packer was lowered into RF1 and placed so that the center of the injection section was at the target depth. The resin fracturing started by expanding the lower packer, and then 1.3 L of resin was filled in the injection interval, which was sealed by expanding the upper packer. The resin was injected with an initial flow rate of 50 mL/min until breakdown, then the flow rate was increased to 100 mL/min, 1 min after breakdown. After the resin fracturing test, a 205 mm diameter core was recovered by drilling co-axially around RF1 (overcoring) to observe resin-filled fractures. Firstly, a large hole (250 mm diameter) was drilled from the ground floor to a depth of 1 m, then. A resin fracturing hole (76 mm diameter) was drilled at the bottom of the 250 mm central hole to a depth of 9 m. Resin fracturing tests were conducted at depths of 2.1 and 3.3 m. After the resin fracturing test, a 205 mm diameter core was recovered by drilling coaxially around the RF1.

Data acquisition methods (microseismic monitoring, fluorescent resin)

The coagulable resin is a type of 2-hydroxyethyl methacrylate (HEMA) based resin that offered a pot-life of more than 2 h and coagulated approximately 18 h after mixing the two liquids. The two liquids (resins) began to coagulate when mixed in a 1:1 volume ratio. The viscosities of both liquids before mixing were six to seven centipoises (cP), and the density after mixing was 1.01 g/cm3. The resin included a fluorescent agent that emitted light under black light. The two liquids were sent separately to a double packer while pressurizing at the same volume rate.

The double packer (Inflatable Packers International Pty Ltd. (IPI)) for the resin fracturing experiments was designed with four lines to flow the liquids, two lines to send the resins, and two lines to send water to expand the upper and lower packers. The lines for the resins are connected to the static mixer outside. The diameter was 70 mm, the injection interval was 475 mm, and the pressure resistance was 50 MPa. A pressure measuring box (OYO Geo-monitoring Service Corporation) was installed in the upper part of the packer to measure the liquid flow pressure and packer pressure.

A total of 24 AE sensors with a resonance frequency of 70 kHz were deployed. Five AE sensors were installed in each of the four AE observation bore holes, and four AE sensors were installed in the remaining one. The total length of a single sensor array was 2 m, with sensors positioned at intervals of 500 mm. Each AE sensor was pushed against the wall of the borehole by a small hydraulic jack set placed behind the sensor. P-wave velocities were assessed before each resin fracturing test. Vibrations were emitted towards each AE sensor in five directions of the AE holes at both depths for the resin fracturing tests. Table 1 lists the conditions used for the AE measurements.

Table 1—Conditions used for the AE measurements. (Source: Reprinted verbatim from Takeuchi et al., 2025)

Item	Condition
Sampling rate	1 MHz
Waveform recording	Continuous recording
Amplification factor setting	Pre-amplifier gain 40 dB Main-amplifier gain 6 dB Total 46 dB
Filter setting	HPF: 20 kHz
	LPF: 500 kHz

Numerical Modeling Approach

Model setup and boundary conditions

The 3D numerical model was constructed to replicate the geological conditions observed at the Kamioka mine field test site. The three-dimensional simulation domain represents a cubic rock mass with dimensions corresponding to the field test scale at the Kamioka site. The computational domain incorporates a central vertical borehole aligned with the injection point, as illustrated in Fig. 1. The model geometry captures the essential geometric features of the field test setup, including the borehole configuration.

The finite element mesh consists of hexahedral elements with systematic refinement around the injection zone to adequately resolve the phase-field interface evolution. The mesh employs a characteristic element size of $h_e = 0.015\ m$ in the refined zone, satisfying the critical relationship $l_s/h_e \geq 2$, where l represents the phase-field regularization parameter ($l_s = 0.03\ m$). This constraint ensures sufficient resolution of the diffuse fracture interface while maintaining computational efficiency for field-scale problems.

Mechanical boundary conditions are applied to simulate hydrostatic pressure conditions at the experimental depth, as shown in Fig. 2. To prevent rigid body motion while allowing fracture-induced deformation, all external boundaries are constrained in their normal directions through Dirichlet boundary conditions:

- The x-positive and x-negative boundaries are fixed in the x-direction
- The y-positive and y-negative boundaries are fixed in the y-direction
- The z-positive and z-negative boundaries are fixed in the z-direction

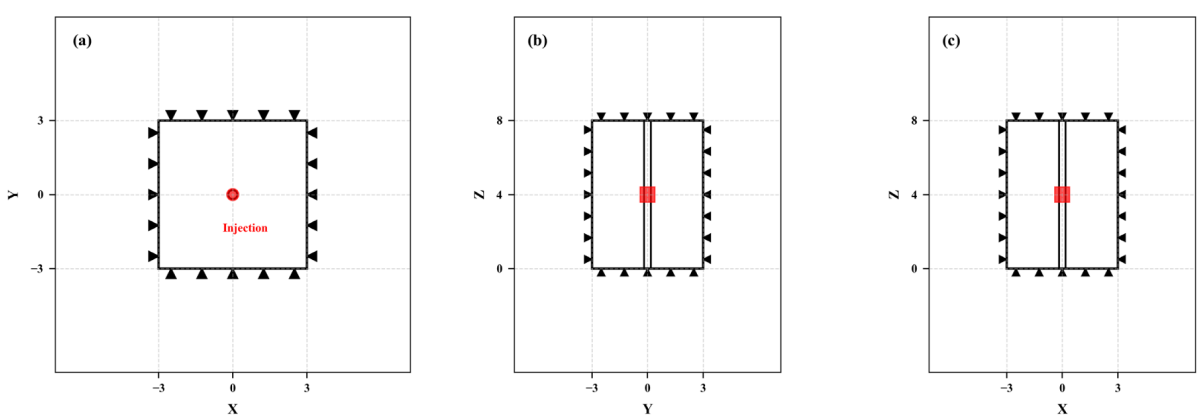

Figure 2—Simulation domain with fixed normal displacement boundary conditions. (a) X-Y view (top); (b) Y-Z view (side); (c) X-Z view (front). The model uses hydrostatic pressure conditions with faces constrained in their normal directions.

This boundary condition approach ensures mechanical stability while allowing unrestricted deformation parallel to each boundary face, enabling natural fracture propagation without pre-imposed stress anisotropy.

Hydraulic boundary conditions establish the fluid pressure environment within the simulation domain. All external boundaries maintain atmospheric pressure conditions ($p = 0\ Pa$) through Dirichlet boundary conditions, representing the connection to the mine environment. The fluid injection is implemented through a volumetric source term applied to the mesh elements surrounding the central borehole. The injection rate follows a time-dependent profile: $Q(t) = 8.333 \times 10^{-8}\ m^3/s$ per unit volume, activated and maintained constant throughout the fracturing process.

Phase-field boundary conditions ensure proper regularization of the fracture evolution process. Dirichlet boundary conditions impose $\upsilon = 1.0$ (intact state) on all external boundaries, preventing artificial fracture initiation at the domain boundaries. This boundary condition maintains the physical constraint that fractures must initiate within the domain rather than propagating from external boundaries.

The initial conditions establish the pre-fracturing state of the system. The displacement field is initialized to zero throughout the domain, while the pressure field begins at atmospheric conditions ($p = 0\ Pa$). The phase-field variable is uniformly initialized to $\upsilon = 1.0$, representing completely intact rock. Pre-existing fracture nucleation sites or initial flaws are not explicitly modeled, allowing the simulation to capture the natural fracture initiation process driven by the hydraulic pressure and local stress concentrations.

This boundary condition framework provides a comprehensive representation of the field test environment while maintaining computational tractability for the complex 3D hydraulic fracturing simulation.

Material properties and parameters

The material properties and numerical parameters used in the simulation are representative of the metamorphic rock mass encountered at the field site. The parameter selection reflects the gneissic geology and the specific experimental conditions of the fluorescent resin injection test. Table 2 summarizes all parameters used in the simulation.

Table 2—Material and numerical properties for the Kamioka field test simulation.

Name	Value	Symbol	Unit
Young's modulus	50.0×10^9	E	Pa
Poisson's ratio	0.33	υ	-
Solid density	2.7×10^3	ρ_s	kg/m^3
Fluid density	1.0×10^3	ρ_f	kg/m^3
Fluid viscosity	1.0×10^{-3}	μ	$Pa\text{-}s$
Fluid compressibility	4.55×10^{-10}	c_f	Pa^{-1}
Biot coefficient	0.3	α_m	-
Matrix permeability	6.0×10^{-10}	K_m	m^2
Porosity	0.0055	φ_m	-
Fracture toughness	10	G_c	N/m

The elastic properties reflect the high stiffness and low porosity characteristic of metamorphic gneiss formations. Young's modulus of 50 GPa is consistent with laboratory measurements on similar crystalline rocks, while Poisson's ratio of 0.33 reflects the felsic composition with significant quartz and feldspar content. The fluid properties correspond to the coagulable fluorescent resin used in the field experiment, with density and viscosity values matching the experimental documentation.

The phase-field formulation employs the AT1 model to handle asymmetric fracture behavior under tensile and compressive loading conditions. The fracture toughness is uniform throughout the intact rock domain, with lower values assigned to natural fracture locations to represent the reduced fracture toughness along pre-existing discontinuities.

The poroelastic coupling parameters reflect the low-permeability, low-porosity nature of the intact rock matrix, ensuring that fluid flow occurs primarily through induced fractures rather than through the rock matrix. This configuration is consistent with the observed experimental behavior and the physical characteristics of competent metamorphic rocks.

Interface Modeling Implementation and Simulation Cases

The inclusion of interface modeling in this study is motivated by field observations from the Kamioka experiment, where Takeuchi et al. (2025) documented that "open natural fractures were developed in the bedrock of the experimental site" and were "observed in the cores recovered during the experiment." Furthermore, acoustic emission analysis revealed that some events were associated with reactivation of existing natural fractures with north-south trending orientations. While the field study did not provide specific geometric details of these natural fractures, their documented presence and influence on fracture propagation behavior necessitate a modeling approach that can capture hydraulic fracture interactions with pre-existing discontinuities.

Weak Interface Representation. The phase-field method's capability to handle complex fracture interactions is leveraged through an interface modeling approach that assigns distinct fracture toughness properties to different material regions. Natural fractures and weak interfaces are represented using the effective interface fracture toughness method, which provides a robust framework for modeling fracture deflection, penetration, and branching behaviors without requiring explicit geometric representation of pre-existing discontinuities.

Based on geological observations from the Kamioka site, a weak interface zone was incorporated into the model with specific geometric characteristics representing the metamorphic foliation structures observed in the field test location. The interface geometry follows the dimensions: DX = 0.8 m, DY = 0.7 m, and DZ = 1.0 m, positioned at a distance of 0.2 m from the wellbore centerline, as illustrated in Fig. 3. This orientation and positioning were selected to represent natural discontinuities at the site. The interface modeling approach utilizes a spatially varying fracture toughness field where the bulk material maintains standard fracture properties while the interface regions are assigned reduced toughness values.

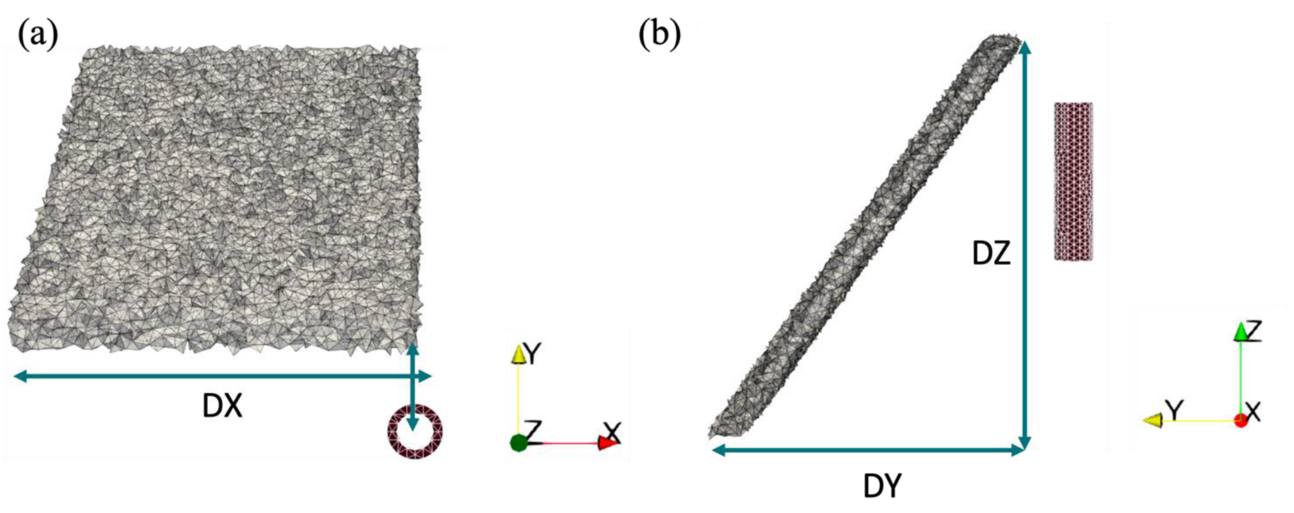

Figure 3—Interface geometry for Cases 2 and 3: (a) X-Y view and (b) Y-Z view showing weak interface dimensions and positioning relative to wellbore.

Simulation Case Definition. Three distinct simulation cases were developed to systematically evaluate the influence of natural fracture strength on hydraulic fracture propagation patterns:

Case 1: Homogeneous Material represents the baseline scenario without any natural fractures or weak interfaces. The entire computational domain maintains uniform fracture toughness properties corresponding to intact rock material (G_c = 10 N/m). This case provides the reference fracture geometry for comparison with heterogeneous scenarios and establishes the fundamental fracture propagation behavior under uniform material properties.

Case 2: Weak Interface ($G_{interface}/G_{bulk}$ = 1/10) incorporates a significantly weakened interface zone where the fracture toughness is reduced to one-tenth of the bulk material value. With G_{bulk} = 10 N/m, the interface fracture toughness becomes $G_{interface}$ = 1 N/m. This substantial reduction in fracture resistance represents scenarios where natural fractures, bedding planes, or foliation planes exhibit markedly lower strength than the surrounding rock matrix. Such conditions are commonly observed in metamorphic terranes where structural anisotropy creates preferential failure paths.

Case 3: Moderate Interface ($G_{interface}/G_{bulk}$ = 1/2) models an intermediate condition where the interface toughness is reduced to half the bulk material value, resulting in $G_{interface}$ = 5 N/m. This configuration represents partially healed natural fractures, cemented joints, or interfaces with moderate strength contrast relative to the bulk rock. This case enables assessment of fracture behavior when natural discontinuities provide some resistance to propagation while still representing zones of relative weakness.

The bulk fracture toughness value of G_{bulk} = 10 N/m was selected based on typical values for metamorphic rocks similar to those encountered at the Kamioka site. This value provides realistic fracture propagation behavior while maintaining numerical stability throughout the simulation duration and enabling convergent solutions across all interface strength scenarios.

Implementation Strategy. The interface zones are implemented through spatial assignment of fracture toughness values within the finite element mesh. Elements located within the predefined interface geometry receive the reduced toughness values, while all other elements maintain the bulk rock properties. This approach ensures that the phase-field evolution accounts for the varying resistance to fracture propagation without requiring complex geometric constraints or interface tracking algorithms.

The transition between bulk and interface properties is smoothed over a characteristic length scale comparable to the phase-field regularization parameter to avoid numerical instabilities associated with sharp material property contrasts. This regularization maintains the physical representation of the interface while ensuring robust numerical performance during fracture propagation.

This systematic approach enables direct comparison of fracture propagation behavior across different interface strength scenarios, providing insights into the mechanisms controlling hydraulic fracture interaction with natural discontinuities in complex geological environments.

Results and Discussion

Summary of Key Simulation Results

Case	Interface Strength	Fracture Geometry	Pressure Response
Case 1	Homogeneous	Radial symmetric propagation	Uniform pressure distribution
Case 2	Weak ($G_{interface}/G_{bulk}$ = 1/10)	Strong deflection along interface, elongated geometry	Enhanced channeling, rapid pressure decline
Case 3	Moderate ($G_{interface}/G_{bulk}$ = 1/2)	Partial interface following, mixed geometry	Intermediate behavior, moderate channeling

Detailed Analysis

The three-dimensional phase-field simulations captured complex hydraulic fracture propagation behavior under different geological conditions, demonstrating the capability to represent non-planar fracture growth and interactions with natural discontinuities observed at the Kamioka site.

Baseline Fracture Behavior (Case 1: Homogeneous Material). Figure 4 presents the phase-field evolution for Case 1, representing homogeneous material without natural fractures or weak interfaces. The fracture initiates around the injection point and propagates radially outward in a symmetric pattern. At t = 3×10^{-4} s (Fig. 4a), initial damage nucleation occurs near the wellbore where hydraulic pressure exceeds material strength. The fracture continues developing (Fig. 4b, t = 8×10^{-4} s) with gradual expansion in multiple directions, maintaining roughly spherical geometry. By t = 1.3×10^{-3} s (Fig. 4c), substantial fracture development is achieved while preserving radial symmetry typical of homogeneous media.

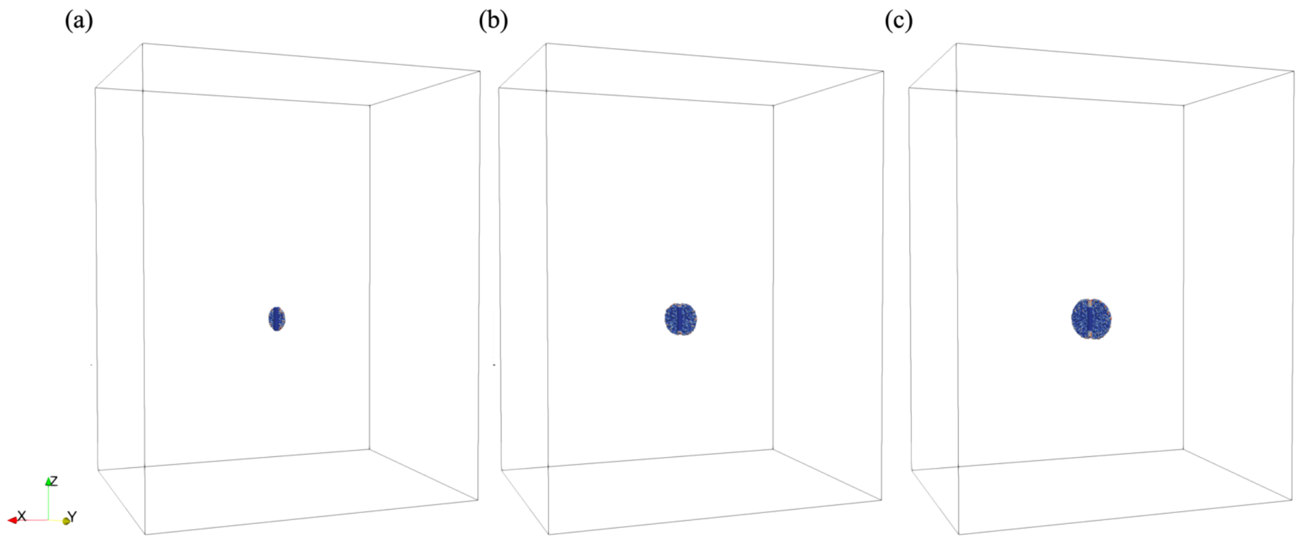

Figure 4—Phase field evolution for Case 1 (without interface): (a) t = 3×10^{-4} s, (b) t = 8×10^{-4} s, (c) t = 1.3×10^{-3} s

This baseline case establishes fundamental fracture behavior under uniform material properties and serves as reference for heterogeneous scenarios. The symmetric propagation reflects natural hydraulic fracture behavior in homogeneous media with uniform boundary conditions, where fractures propagate radially outward from the injection point without preferential directions imposed by material heterogeneities.

Weak Interface Influence (Case 2). The introduction of weak interfaces with $G_{interface}/G_{bulk}$ = 1/10 dramatically alters fracture propagation behavior (Fig. 5). The phase-field evolution reveals clear deflection toward the weakened zone, demonstrating preferential propagation along reduced fracture toughness zones. At t = 3×10^{-4} s (Fig. 5a), fracture initiation follows similar patterns to the homogeneous case, but subsequent development shows marked asymmetry as the fracture encounters the interface region.

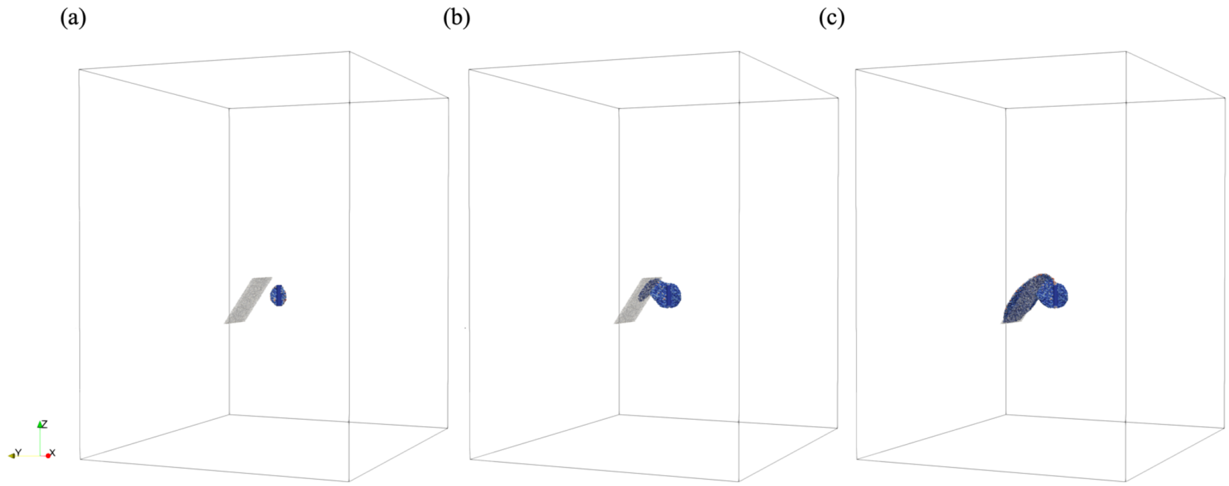

Figure 5—Phase field evolution for Case 2 (weak interface): (a) t = 3×10⁻⁴ s, (b) t = 8×10⁻⁴ s, (c) t = 3.5×10⁻³ s

Fracture deflection becomes pronounced at $t = 8\times10^{-4}$ s (Fig. 5b), where propagating fractures clearly follow weak interface geometry rather than maintaining radial symmetry. This behavior continues through $t = 3.5\times10^{-3}$ s (Fig. 5c), resulting in elongated fractures aligned with interface orientation. The simulation captures competition between hydraulic pressure and material heterogeneity, showing similarity to field observations from the Kamioka experiment, where Takeuchi et al. (2025) documented hydraulic fracture interactions with pre-existing natural fractures and metamorphic foliation structures. Specifically, at 3.30 m depth in the field experiment, bi-wing vertical fractures extended in the NE-SW direction and changed direction when encountering natural fractures. The preferential propagation along weak interfaces reflects geological discontinuity influences commonly encountered in crystalline rock masses.

Moderate Interface Effects (Case 3). Case 3, featuring moderate interface strength contrast ($G_{interface}/G_{bulk} = 1/2$), produces fracture behavior intermediate between homogeneous and weak interface scenarios (Fig. 6). The fracture evolution shows some deflection toward interfaces while maintaining more symmetric characteristics compared to Case 2. At $t = 6\times10^{-4}$ s (Fig. 6a), initial development appears similar to baseline cases, but subtle asymmetries emerge as fractures approach interface zones.

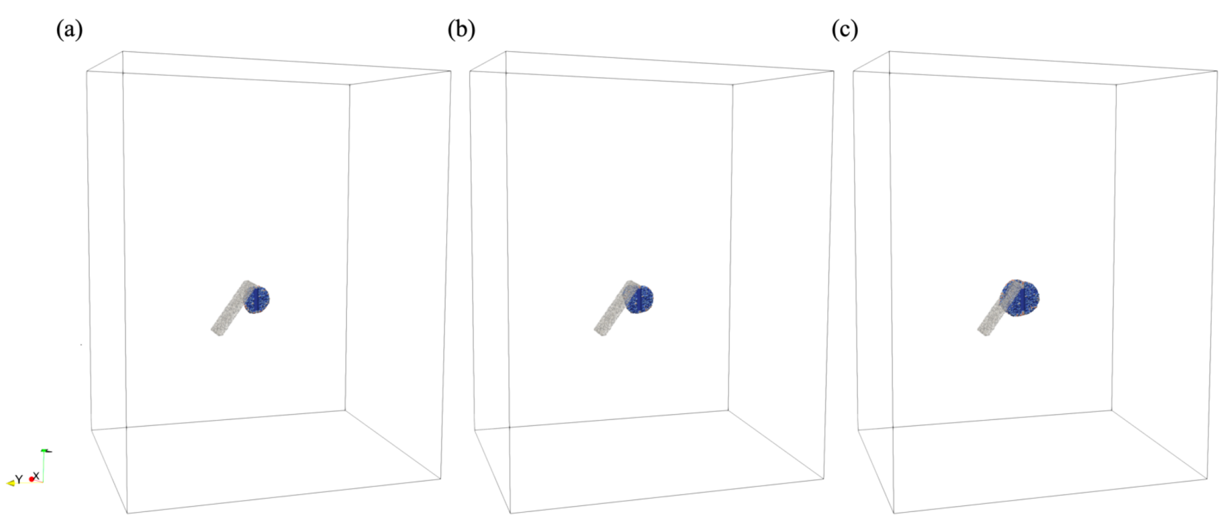

Figure 6—Phase field evolution for Case 3 (moderate interface): (a) t = 6×10⁻⁴ s, (b) t = 7×10⁻⁴ s, (c) t = 1.2×10⁻³ s

The moderate strength contrast allows fractures to partially follow interfaces while propagating through surrounding bulk material (Fig. 6b, t = 7×10^{-4} s). By t = 1.2×10^{-3} s (Fig. 6c), the final fracture geometry exhibits mixed characteristics, combining radial propagation and interface-controlled growth elements.

Pressure Field Analysis. The pressure field distributions (Fig. 7) reveal significant differences in fluid flow patterns between weak and moderate interface cases. For weak interfaces (Fig. 7a), pressure fields exhibit strong channeling along fracture pathways, with enhanced pressure propagation through preferential flow routes created by deflected fractures. The pressure distribution shows clear anisotropy, reflecting non-uniform permeability fields created by fracture networks.

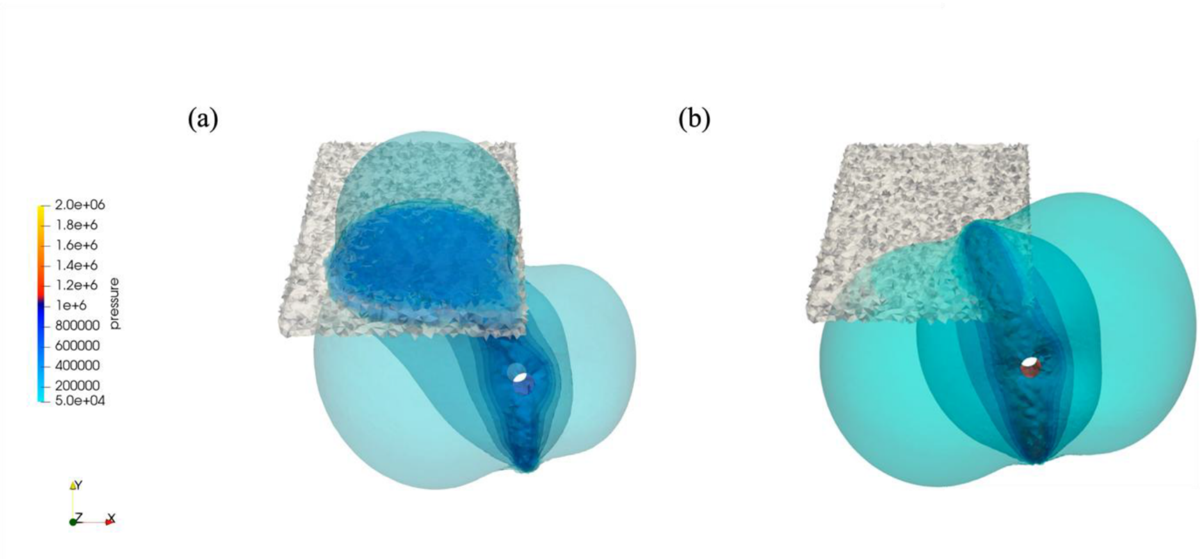

Figure 7—Pressure field distribution comparison: (a) weak interface (Case 2) vs (b) moderate interface (Case 3)

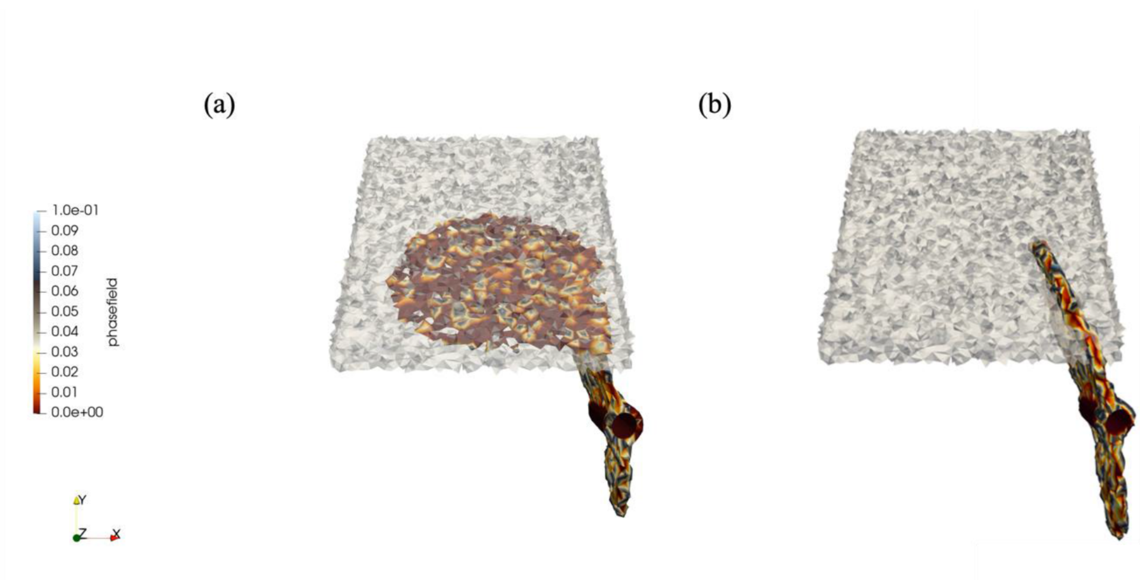

Figure 8—Phase field damage distribution comparison: (a) weak interface (Case 2) vs (b) moderate interface (Case 3)

In contrast, moderate interface cases (Fig. 7b) display more distributed pressure fields with less pronounced channeling effects. Pressure distributions maintain greater symmetry while showing interface influence, indicating more balanced flow between fracture networks and the surrounding matrix. These pressure behavior differences have important implications for injection efficiency and fracture containment.

The pressure evolution during hydraulic fracturing (Fig. 9) demonstrates distinct behaviors for cases with and without weak interface modeling. Weak interface cases show characteristic pressure peaks followed by more rapid decline, reflecting enhanced connectivity created by fracture deflection along interfaces. This pressure response pattern aligns with breakdown behavior observed in naturally fractured formations, where preferential flow paths lead to pressure communication over larger distances.

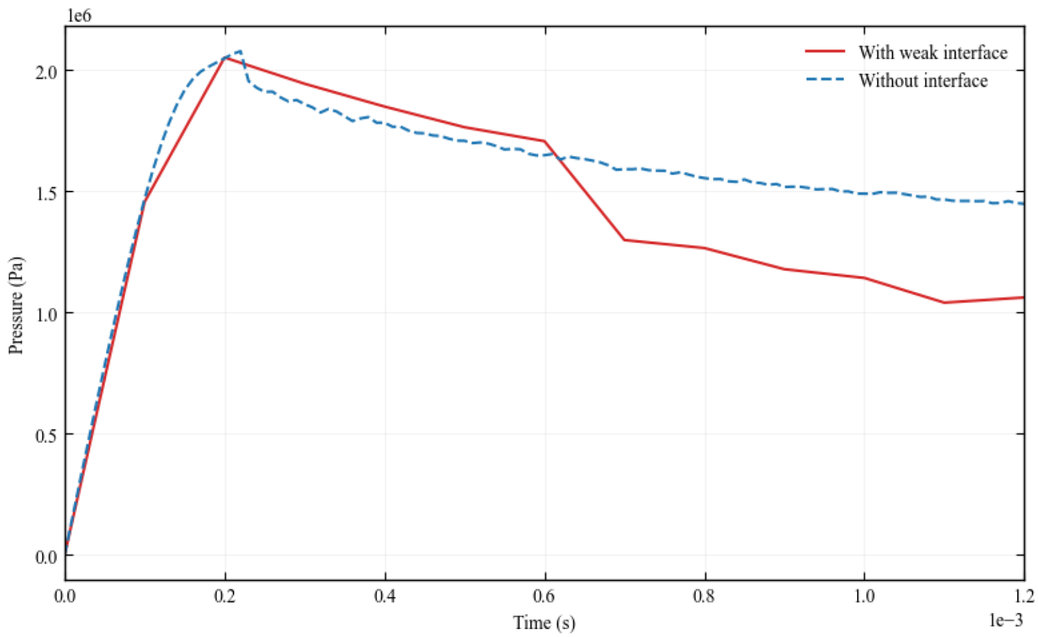

Figure 9—Pressure evolution during hydraulic fracturing with and without weak interface modeling

The pressure evolution trends observed in the simulation show qualitative similarity to the field experiment behavior, where breakdown pressures varied with depth (5.86 MPa at 2.10 m and 9.50 MPa at 3.30 m) and interface strength. The lower breakdown pressure at shallow depth reflects natural fracture influence and reduced confining stress, consistent with the enhanced pressure communication observed in weak interface scenarios.

Damage Distribution Comparison. Figure 8 provides a detailed comparison of phase-field damage distribution between weak and moderate interface cases. The weak interface case (Fig. 8a) shows extensive damage development within interface zones, with complex fracture networks exhibiting interface-controlled propagation. The damage pattern reveals how fractures interact with heterogeneities to create connected pathways extending beyond immediate injection zones.

The moderate interface case (Fig. 8b) displays more localized damage with less extensive interface activation. Fractures maintain a more compact geometry through subtle deflection patterns. This damage distribution difference directly impacts stimulated reservoir volume and has practical implications for completion design optimization.

The complex fracture geometries predicted by the simulations show consistency with field observations from fluorescent resin visualization under black light, which provided direct evidence of non-planar fracture propagation behavior in the recovered cores. The preferential flow pathways created by interface-controlled deflection match the resin-filled fracture networks documented in the field experiment.

Conclusions

This study presents a high-fidelity 3D hydraulic fracture model based on the phase-field approach through simulation, following the Kamioka mine experiment. The research demonstrates the capability of phase-field methods to capture complex fracture-interface interactions in heterogeneous geological environments, providing valuable insights for hydraulic fracturing operations.

The systematic comparison of three simulation cases reveals fundamental insights into fracture propagation mechanisms in heterogeneous media. The transition from symmetric radial growth (Case 1) to interface-controlled elongated geometries (Case 2) demonstrates how geological heterogeneities dominate fracture development under moderate strength contrasts. Fracture toughness ratios between interfaces and bulk material serve as critical parameters controlling fracture trajectory, with 10:1 strength contrasts producing dramatic deflection while 2:1 contrasts yield subtle but significant effects.

The phase-field approach proves particularly effective at capturing complex interactions without requiring explicit interface tracking or specialized meshing procedures, enabling natural representation of fracture deflection, branching, and arrest phenomena that conventional methods struggle to handle efficiently.

This work provides a valuable benchmark case for hydraulic fracture modeling in complex geological environments. Future work should focus on quantitative validation and detailed acoustic emission comparison, including first P-wave motion analysis as demonstrated in the Kamioka experiment, and explicit modeling of shear slip mechanisms along pre-existing interfaces. The integration of shear slip modeling with phase-field damage evolution would provide a more comprehensive representation of mixed-mode failure behavior observed in naturally fractured formations.

Nomenclature

Roman Symbols

$\boldsymbol{u}$	displacement vector, *m*
p	fluid pressure, *Pa*
$\boldsymbol{b}$	body force vector, N/m^3
$\bar{\boldsymbol{q}}$	prescribed fluid flux, *m/s*
E	Young's modulus, *Pa*
h_e	characteristic finite element size, *m*
t	time, *s*
Δt	time step size, *s*
$\underline{G_c}$	critical energy release rate (fracture toughness), *N/m*
G_{bulk}	fracture toughness of bulk rock material, *N/m*
$G_{interface}$	fracture toughness of interface material, *N/m*
K_m	matrix permeability, m^2
c_f	fluid compressibility, Pa^{-1}
C_{eff}	effective stiffness tensor degraded by phase-field variable, *Pa*
$\mathbf{I}$	identity tensor, dimensionless
$g(v)$	degradation function for phase-field model, dimensionless
$M(v)$	Biot modulus, *Pa*
Q	volumetric source term, m^3/s per unit volume
DX, DY, DZ	interface dimensions in x, y, z directions, *m*
l_s	phase-field regularization length parameter, *m*
Tr	trace operator, dimensionless

Greek Symbols

υ phase-field variable (0 = fully fractured, 1 = intact), dimensionless
$\boldsymbol{\sigma}$ total stress tensor, *Pa*
$\boldsymbol{\varepsilon}$ strain tensor, dimensionless
Ω computational domain, m^3
Γ fracture set, m^2
ν Poisson's ratio, dimensionless
ρ_f fluid density, kg/m^3
ρ_s solid density, kg/m^3
μ fluid dynamic viscosity, $Pa{\cdot}s$
α_m Biot coefficient, dimensionless
φ_m matrix porosity, dimensionless
ψ strain energy density, J/m^3
k_{eff} effective bulk modulus degraded by phase-field variable, *Pa*

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